

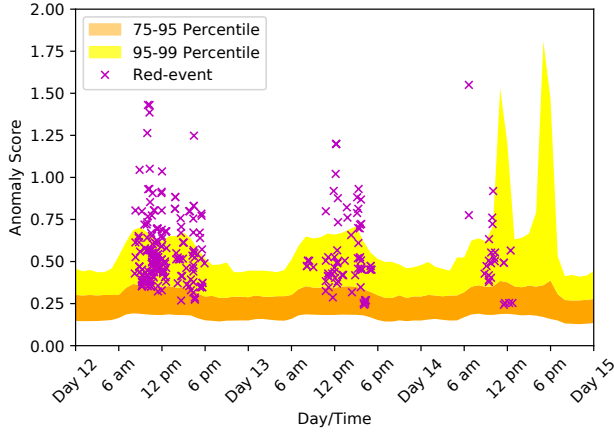


Figure 6: Character-level red-team log-line anomaly scores in relation to percentiles over time.

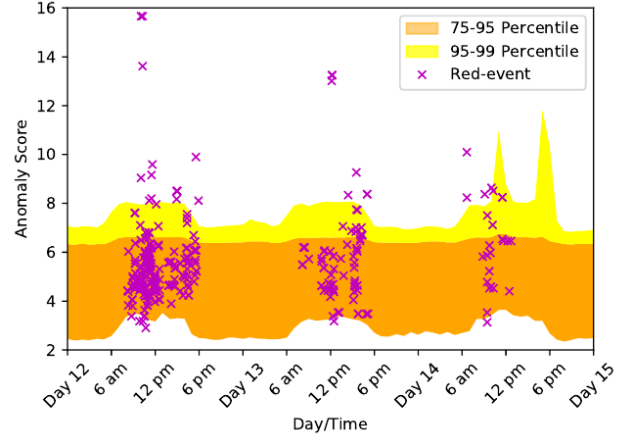


Figure 7: Word-level red-team log-line anomaly scores in relation to percentiles over time.

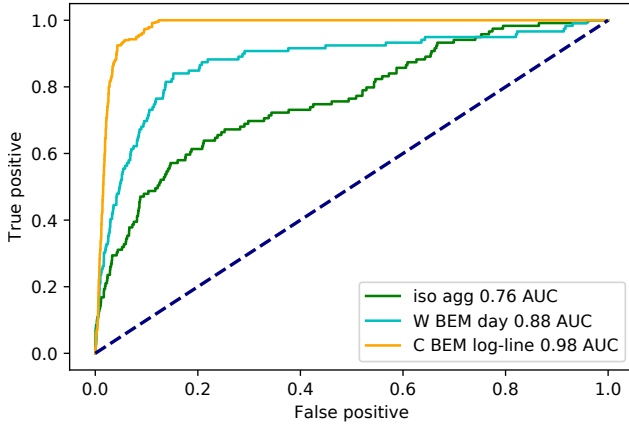


Figure 8: ROC curves for best performing baseline, word language model evaluated at day-granularity, and character language model evaluated at log-line-granularity.

ularity, and the event-level granularity.

Since the proportion of red-team to normal events is vanishingly low in the data-set ($< 0.001\%$), the false-positive rate is effectively the proportion of data flagged to achieve a particular recall. From this observation, Figure 8 shows the character event model can achieve 100% recall from only 12% of the data whereas the other models considered only achieve 100% recall when nearly all of the data has been handed to the analyst. Further, the character event model can achieve 80% recall by flagging only 3% of the data whereas the word day language model needs 14% of the data and the aggregate isolation forest model needs 55% of the data to achieve the same result.

5 Conclusion

This work builds upon advances in language modeling to address computer security log analysis, proposing an unsupervised, online anomaly detection approach. We eliminate

the usual effort-intensive feature engineering stage, making our approach fast to deploy and agnostic to the system configuration and monitoring tools. It further confers the key advantage of event-level detection which allows for a near immediate alert response following anomalous activity.

In experiments using the Los Alamos National Laboratory Cyber Security Dataset, bidirectional language models significantly outperformed standard methods at day-level detection. The best log-line-level detection performance was achieved with a bidirectional character-based language model, obtaining a 0.98 area under the ROC curve, showing that for the constrained language domain of network logs, character based language modeling can achieve comparable accuracy to word based modeling for event level detection. We have therefore demonstrated a simple and effective approach to modeling dynamic networks with open vocabulary logs (e.g. with new users, PCs, or IP addresses).

We propose to extend this work in several ways. First, potential modeling advantages of tiered architectures merit further investigation. The use of tiered architectures to track PCs instead of network users, or from a richer set of logging sources other than simply authentication logs may take better advantage of their modeling power. Next, we anticipate interpretability can become lost with such detailed granularity provided by log-line-level detection from a character-based model, therefore future work will explore alternate methods of providing context to an analyst. Finally, we are interested in exploring the robustness of this approach to adversarial tampering. Similarly performing models could have different levels of resilience that would lead to selection of one over another.

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X and Titan Xp GPUs used in this research.

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